# Original Articles

# Aspartame Use By Persons with Diabetes

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Sixty-two subjects having either insulin-dependent or non-insulin-dependent diabetes completed a randomized, double-blind study comparing effects of aspartame or a placebo on blood glucose control. Twenty-nine subjects consumed 2.7 g aspartame per day for 18 wk, given as aspartame-containing capsules with meals, while 33 subjects took identical appearing placebo capsules. After 18 wk, no changes were seen in fasting or 2-h postprandial blood glucose levels or glycohemoglobin levels in either the aspartame- or placebo-treated groups. Adverse reactions were no more common in the group taking aspartame. We conclude that use of aspartame as a low-calorie sweetener does not adversely affect glycemic control of persons with diabetes. DIABETES CARE 1985; 8:415–17.

Although there is no absolute requirement for sweeteners in the diet, the availability of low-calorie, noncarbohydrate sweeteners is believed to make it easier for diabetic individuals to adhere to meal plans. In obese subjects, replacement of sucrose with aspartame has resulted in a reduction of caloric intake.[1] In surveys of persons with non-insulin-dependent diabetes mellitus (NIDDM) and of parents of children with insulin-dependent diabetes mellitus (IDDM), the majority found the availability of artificially sweetened foods to be useful in maintaining an appropriate diet.[2,3] In the past decade, however, the safety of two common non-nutritive sweeteners, saccharin and cyclamate, has been questioned. Furthermore, apart from safety considerations, saccharin has an unpleasant aftertaste that makes it an undesirable sweetener for many persons. Therefore, many individuals with diabetes have welcomed the availability of aspartame as an alternative, low-calorie sweetener.

Aspartame (Nutrasweet, G.D. Searle & Co., Skokie, Illinois), is a dipeptide containing aspartic acid and phenylalanine methyl ester. Numerous studies have demonstrated its safety in adults and children over the age of 2 yr, as well as in lactating women and phenylketonuria (PKU) heterozygotes. These studies have been reviewed recently.[4,5] Aspartame is now increasingly being incorporated into meal plans for diabetic individuals. However, to our knowledge, there has been only one study examining what effect aspartame might have on diabetes control.[6] Therefore, we have done an 18-wk randomized, placebo-controlled, double-blind study in which aspartame was given to persons with IDDM and NIDDM. The amount given, 2.7 g per day (equivalent to 40 mg/kg in a 70-kg person), is somewhat higher than that likely to be ingested by 99% of the population when aspartame replaces dietary sucrose.[7] We estimate that most individuals will not consume more than 0.5 g of aspartame per day.[5]

Recently, several anecdotal and undocumented reports in the lay press have suggested a number of adverse reactions associated with use of aspartame. These include headaches and other neurologic symptoms. The use of a placebo-controlled, double-blind experimental design allowed us to evaluate the significance of these reports.

## MATERIALS AND METHODS

*Study population.* Subjects were at least 18 yr old and not >65 yr of age by the beginning of the study. They were assigned to groups as follows: (1) insulin-dependent diabetes mellitus (IDDM): subjects who, by history, developed ketosis or ketoacidosis when adequate exogenous insulin was not provided and (2) non-insulin-dependent diabetes mellitus (NIDDM): subjects who are not on insulin and are not ketotic or who, if on insulin, have no history of ketoacidosis.

For both groups, the diagnosis of diabetes had been established by a fasting plasma glucose >140 mg/dl, an abnormal oral glucose tolerance test as interpreted by the U.S. Public Health Service criteria, or an unequivocal history of diabetes. At the time of entry into the study, fasting plasma glucose

TABLE 1
Fasting plasma glucose (mg/dl ± SE)

| | Pretreatment | | During treatment | | |
|---|---|---|---|---|---|
| | Wk −1 | Wk 0 | Wk 9 | Wk 17 | Wk 18 |
| Placebo | 187 ± 15 | 177 ± 24 | 175 ± 16 | 173 ± 13 | 168 ± 14 |
| Aspartame | 186 ± 11 | 178 ± 14 | 200 ± 15 | 195 ± 18 | 174 ± 14 |

TABLE 3
Glycohemoglobin (% ± SE)

| | Pretreatment | | During treatment | | |
|---|---|---|---|---|---|
| | Wk −1 | Wk 0 | Wk 9 | Wk 17 | Wk 18 |
| Placebo | 10.7 ± 0.4 | 10.8 ± 0.5 | 10.7 ± 0.5 | 11.0 ± 0.5 | 11.4 ± 0.6 |
| Aspartame | 12.0 ± 0.6 | 11.8 ± 0.5 | 11.5 ± 0.4 | 11.3 ± 0.4 | 11.4 ± 0.5 |

was not >200 mg/dl and therapy had been stable (no change of medication or dosage) for at least 1 mo. If glycemic control changed during the study, subjects were not dropped from the study. Subjects were recruited from clinics of the University of Illinois Hospital, and included persons from diverse ethnic, racial, and socioeconomic backgrounds. After recruitment, subjects were required to give written consent as required by the University Institutional Review Committee. Subjects were then assigned to a randomization schedule depending on their type of diabetes. There were 32 subjects with IDDM and 31 with NIDDM.

*Test substances.* Aspartame was given in the form of capsules, each containing 0.3 g aspartame. Three capsules were taken with each meal, for a total of 9 capsules per day (2.7 g aspartame). The placebo consisted of identical capsules filled with 0.2 g corn starch. This amount of starch (1.8 g/day) is unlikely to have affected diabetes control.

Capsules were provided in coded bottles, which contained either aspartame or placebo according to a randomization table, and were assigned to subjects in sequential order. Separate sequences were used for IDDM and NIDDM, to insure equal numbers in each group, and randomization was further blocked by weight so that there would be approximately equal numbers of nonobese (<115% ideal body weight, IBW), moderately obese (115–130% IBW), and very obese (>130% IBW) subjects on aspartame and placebo. There were no breaks in the double-blind code during the study.

*Experimental design.* Each subject had two baseline visits 1 wk apart, at which blood was drawn for fasting and 2-h postprandial glucose values and glycohemoglobin levels. Subjects fasted overnight before coming for the initial blood drawing. They then had their breakfast (usually in the hospital cafeteria) and returned 2 h after the start of breakfast for the second blood drawing. Subjects started taking aspartame or placebo capsules after the second set of baseline blood samples were drawn. During the study, each subject kept a log of capsules taken. Subjects were asked to make up missed or forgotten capsules with the next meal, or as soon as they remembered. Return visits were scheduled after 3, 6, 9, 12, 15, 17, and 18 wk of taking the capsules. At each visit, the logs of capsules ingested were collected, the bottles of capsules collected and remaining capsules counted, and a new bottle of capsules disbursed. Nondirective questions were asked regarding possible adverse reactions as well as intercurrent illnesses. In addition, at 9, 17, and 18 wk, blood samples were drawn as in the baseline period. The fasting samples were drawn before the capsules were taken, and the postprandial samples approximately 2 h after taking the capsules.

*Analytic methods.* Glucose was determined using a Yellow Springs Glucose Analyzer (Yellow Springs Instruments, Yellow Springs, Ohio). Glycohemoglobin was determined electrophoretically using the Corning method (Corning Instruments, Palo Alto, California).

TABLE 2
Two-hour postprandial plasma glucose (mg/dl ± SE)

| | Pretreatment | | During treatment | | |
|---|---|---|---|---|---|
| | Wk −1 | Wk 0 | Wk 9 | Wk 17 | Wk 18 |
| Placebo | 227 ± 18 | 233 ± 18 | 226 ± 23 | 220 ± 16 | 245 ± 20 |
| Aspartame | 235 ± 20 | 241 ± 21 | 268 ± 20 | 242 ± 16 | 244 ± 19 |

## RESULTS

Of the 63 subjects, 62 completed the study. One (discussed below) dropped out because of gastrointestinal symptoms. There were no serious instances of noncompliance, although a number of subjects forgot occasional doses and later made up the missing doses. Of the 62 subjects completing the study, 29 were on aspartame (15 IDDM and 14 NIDDM) and 33 were on placebo (16 IDDM and 17 NIDDM).

TABLE 4
Adverse reactions

| Aspartame group | Placebo group |
|---|---|
| Headache | Eczema |
| Constipation (two cases) | Dizziness |
| Gastroenteritis | Eye twitching |
| Severe diarrhea | Blurred vision |
| Itch | Foot pain |
| Sinus congestion | Ketoacidosis |
| | Musculo-skeletal pain |
| | Rash (two cases) |
| | Itching (two cases) |
| | Nausea (three cases) |
| | Loose stools |
| | Less frequent stools |
| | Constipation |
| | Gastroenteritis |
| | Diarrhea |
| | Malaise |
| | Dry skin |

Because there were no differences in mean changes in plasma glucose or glycohemoglobin levels between subjects with IDDM and NIDDM, these data were pooled and are presented in Tables 1, 2, and 3. There were no significant differences between baseline values in placebo and aspartame groups; also values did not change significantly during 18 wk of treatment. The statistical power of the methodology is such that the probability of falsely accepting the null hypothesis in analyzing the glucose results is <11%.

In the aspartame group, six subjects experienced a total of seven adverse reactions, consisting of headaches, constipation (two cases), itching, sinus congestion, gastroenteritis, and severe diarrhea (Table 4). The diarrhea stopped on discontinuation of the aspartame. This subject was dropped from the study and subsequently rechallenged with the aspartame capsules, and again had diarrhea.

In the placebo group, adverse reactions were experienced in 14 of the subjects. Because of multiple adverse reactions in several subjects, a total of 21 adverse reactions were seen in this group, consisting of eczema, dizziness, eye twitching, blurred vision, foot pain, nausea (3 cases), musculo-skeletal pain, rash (2 cases), itching (2 cases), ketoacidosis, diarrhea, loose stools, less frequent stools, constipation, general malaise, and dry skin.

A number of intercurrent events occurred that were not counted as adverse experiences. In the aspartame group these included acne, upper respiratory infection (four cases), and tenosynovitis. In the placebo group, the following events occurred: oral infection after oral surgery (one case) or from biting the buccal mucosa (one case), upper respiratory infection (six cases), back infection, and right ear discomfort.

## DISCUSSION

The duration of these studies, about 4 mo, was chosen so that any changes in glycemic control might be reflected in altered glycohemoglobin levels. The failure of these values (or of fasting and postprandial plasma glucose values) to change indicates that aspartame, in the highest amounts likely to be ingested, will not lead to changes in glycemic control in IDDM or NIDDM.

When studied in a double-blind protocol, adverse reactions reported on aspartame were no greater than those reported by subjects taking a placebo, although in one subject diarrhea seemed to be specifically related to aspartame ingestion. Because only 29 subjects received aspartame, the incidence of infrequently occurring adverse reactions cannot be estimated from this study. However, the data do confirm previous information on the safety of aspartame, and indicate that adverse reactions are unlikely to occur in persons using aspartame.

Because aspartame was given as capsules, and not actually used as a sweetener, this study gives no information on patient acceptability of aspartame as a sweetener and does not provide data on whether use of aspartame is likely to alter caloric intake, or whether aspartame may reduce plasma glucose levels by reducing intake of sucrose or glucose. However, in conjunction with previous studies demonstrating the safety of aspartame, this investigation confirms that aspartame is safe for use by persons with diabetes.

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